

Force permissions checks on keyfiles and directories
StrictModes yes

At this point, the SSH service has been secured to a great extent. However, it is very essential to configure the monitoring and logging of the service, which gives the administrators a view of the operations of the SSHD service. The following settings are self-explanatory and should be tweaked as per requirements. The log level can be tuned from *Info to Debug 1,2,3* in order to debug a particular SSHD-service-related problem or intrusion attack.

```
# Syslog Logging
SyslogFacility AUTH
# set log level
LogLevel INFO
# Print most recent user login time
PrintLastLog yes
```

Besides the above settings, there are configuration tweaks available for advanced SSH setups such as key-based authentication, password-based authentication, etc. It is important to note that while most of the SSHD settings follow standards, a few distros follow different

configuration keywords and the help manual should be referred to, in such cases.

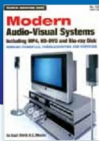
As seen earlier, while the SSH service offers many important settings by default, these may not be enough for advanced IT infrastructures. Introducing two-factor authentication, host-based private and public key-pair-based security, and higher encryption standards for data in transit between server and client, are a few examples of the security mechanisms to be considered. Having a firewall and an intrusion prevention system is also advised for mission-critical data centres running SSH service on servers that carry important business data or applications. Open source utilities such as *fail2ban* or *DenyHosts* should be used to deal with SSH brute force attacks. Regular operating system level patching and SSH service version patching is extremely important for maximum security. 

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